Face Mask Detection System

Project Report — Computer Vision Course (BYOP)

1. Problem Statement

During and after the COVID-19 pandemic, ensuring people wear face masks in public spaces became critical for public health. Manually monitoring mask compliance in crowded areas is impractical. This project aims to build an automated Face Mask Detection System using Computer Vision techniques that can identify whether a person is wearing a face mask or not from an image.

2. Why This Problem Matters

- Public health and safety is a real-world concern
- Manual monitoring is slow, expensive, and error-prone
- An automated system can work in real-time at scale
- Directly applicable in hospitals, schools, offices, and public transport

3. Approach & Solution

The solution was built in three stages:

Stage 1 — Face Detection Used OpenCV's deep neural network (DNN) module with a pre-trained Caffe model (res10_300x300_ssd_iter_140000) to detect faces in images. This model uses Single Shot Detector (SSD) architecture and returns bounding boxes with confidence scores.

Stage 2 — Mask Classification For each detected face, the lower half of the face region was extracted (since masks cover the nose and mouth). HSV color thresholding was applied to detect skin-colored pixels. If the lower face shows little skin (skin ratio < 0.25), it is classified as "Mask", otherwise "No Mask".

Stage 3 — Visualization Results are displayed with colored bounding boxes (green = Mask, red = No Mask), confidence scores, and a summary statistics bar showing total faces, mask count, and compliance rate.

4. Computer Vision Concepts Used

- Image reading and color space conversion (BGR → RGB, RGB → HSV)
- Deep Neural Networks for face detection (cv2.dnn)

- Blob creation for neural network input
- Region of Interest (ROI) extraction
- Color thresholding and masking
- Drawing bounding boxes and text overlays

5. Key Decisions Made

- Used a pre-trained model instead of training from scratch due to time and resource constraints
- Focused on the lower half of the face for mask detection since that is where masks are worn
- Chose HSV color space over RGB because HSV is more robust to lighting changes

6. Challenges Faced

- URLs for test images were unreliable, solved by uploading images directly
- Dark colored masks (black) were harder to detect using color thresholding since they don't show skin color difference
- Colab session resets caused functions to become undefined, solved by restructuring the notebook

7. Limitations

- Color-based mask detection is not robust — dark masks are often misclassified
- The system works on static images, not real-time video
- Performance drops with faces at angles or in low lighting
- A deep learning classifier trained on mask/no-mask dataset would significantly improve accuracy

8. What I Learned

- How to use OpenCV's DNN module with pre-trained models
- How HSV color space works and why it is preferred for color detection
- How face detection pipelines work in practice
- The difference between rule-based approaches and learning-based approaches
- Importance of testing with diverse images to find edge cases

9. Future Improvements

- Train a proper CNN classifier on a labeled mask dataset
- Extend to real-time video detection using webcam
- Add support for incorrectly worn masks (mask below nose)
- Deploy as a web application using Flask or Streamlit